Celia R. Mulcahey

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ORCID: 0000-0003-2434-2657
research interests: Galaxy evolution; Galaxy kinematics; Large-scale structure of the universe

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Education

Johns Hopkins University, Baltimore, Maryland
Ph.D. Candidate in Astronomy & Astrophysics, August 2022–present.
M.A., Physics, May 2024.

Mount Holyoke College, South Hadley, Massachusetts
B.A., Astronomy & Geology, *summa cum laude*, 2021.
Thesis: "Star Formation and AGN Feedback in the Local Universe: Combining the Radio Continuum and Integral Field Spectroscopy."
Advised by Dr. Sarah K. Leslie and Prof. Jason E. Young.

Research positions

Space Telescope Science Institute / Johns Hopkins University
Graduate Research Assistant, August 2022–present.
Advised by Dr. Susan Kassin and Dr. Timothy Heckman.

Maria Mitchell Observatory
Post-Baccalaureate Research Fellow, 2021–2022.
Fast and Fortunate for Fast Radio Burst Follow-up collaboration.
Advised by Dr. Regina Jorgenson and Dr. J. Xavier Prochaska.

Leiden / European Space Agency Astrophysics Program for Summer Students
Research Intern, Summer 2020.
Advised by Dr. Sarah K. Leslie.

Maria Mitchell Observatory
NSF REU Intern, Summer 2019.
Advised by Dr. Laura Prichard and Dr. Regina Jorgenson.

Mount Holyoke College, Mineral Spectroscopy Lab
Research Intern, Summer 2018.
Advised by Dr. M. Darby Dyar.

Talks and posters

Seminars
The University of California, Santa Cruz CGI (Cosmology/Galaxies/IGM) Zoom seminar, October 2025.

Contributed Talks
2026 UCSC Galaxy Workshop, Santa Cruz, CA, August 2026.

Beyond Hubble: Revisiting the Hubble Sequence Across Cosmic Time, Cambridge, UK, April 2026.
Towards the Full Bloom of First Star, Galaxy, and Black Hole Formation Exploration, Tokyo, JP, March 2026.
2025 UCSC Galaxy Workshop, Santa Cruz, CA, August 2025.
Dancing in the Dark: When Galaxies Shape Galaxies, Sexten, IT, June 2025.
2024 UCSC Galaxy Workshop, Santa Cruz, CA, August 2024.

Contributed posters
The Metal-Poor Frontier: Understanding Low-Metallicity Stars and Galaxies Across Cosmic Time, Baltimore, MD, May 2026.
2nd IAA-CSIC Severo Ochoa Advanced School on Galaxy Evolution, Grenada, SP, May 2025.
MIT JWST First Light Conference, Cambridge, MA, June 2023.
237th Meeting of the American Astronomical Society, January 2021.
235th Meeting of the American Astronomical Society, Honolulu, HI, January 2020.

Grants and awards

Selected programs as Principal Investigator
Keck/OH-Suppressing Infrared Integral Field Spectrograph (OSIRIS), 2022.
"Keck/OSIRIS Observations of Globular Cluster [PR95] 30244 in M81." Target of Opportunity Program, allocated 1h.

Selected programs as Co-Investigator (* indicates significant contribution at all stages)
ALMA, Cycle 13, Program 2026.1.00700.S, 2026. P.I.: S. Fujimoto, allocated 30h.
ALMA, Cycle 12, Program 2025.1.00363.S, 2025. P.I.: S. Fujimoto, allocated 40h.
James Webb Space Telescope, Cycle 4, Program 6796, 2025. P.I.: S. Fujimoto, allocated 60.7h.
James Webb Space Telescope, Cycle 2, Program 4291, 2023. P.I.: S. A. Kassin, allocated 67.8h.*
James Webb Space Telescope, Cycle 1, Program 2123, 2021. P.I.: S. A. Kassin, allocated 74.3h.*
Institut de Radioastronomie Millimétrique, Program 171-20, 2020. P.I.: S. K. Leslie, allocated 22h.*

Research and travel grants
Mount Holyoke College Lynk Funding, awarded Spring 2020.
Massachusetts Space Grant, awarded 2018 and 2019.
Mount Holyoke College Student Leadership Center Travel Grant, awarded Fall 2019.
Kenan Institute of Engineering, Technology and Space Science Grant, awarded Fall 2018.

Publications

First-author publications
2. Mulcahey, C. R. *et al.* "Star Formation and AGN Feedback in the Local Universe: Combining LOFAR and MaNGA." A&A, 665, A144 (2022).

1. Mulcahey, C. R. *et al.* "Deciphering the Origin of Ionized Gas in IC 1459 with VLT/MUSE." MNRAS, 504, 4 (2021).

Co-author publication (* indicates significant contribution)
2. Stanghellini, L. *et al.* "Direct Abundance Maps and Radial Metallicity Gradients of two Galaxies at z 4-5 in the GARDEN Survey" eprint arXiv:2601.17148, (2026). *
1. Abdurro'uf, A. *et al.* "Spatially Resolved Stellar Populations of $0.3 < z < 6.0$ Galaxies in WHL 0137-08 and MACS 0647+70 Clusters as Revealed by JWST." ApJ, (2023).

Academic honors

NSF-GRFP, Honorable Mention, awarded Spring 2022 and 2024.
Phi Beta Kappa, awarded Spring 2021.
Oldest and most prestigious academic honor society in the United States.
Mary Dailey Irvine Prize, awarded Spring 2021.
Most distinguished senior thesis in astronomy and excellence in undergraduate research. Awarded by the Five College Astronomy Department.
Bennett Prize, awarded Spring 2018.
Excellence in undergraduate physics. Awarded by the Mount Holyoke College Physics Department.

Teaching and service

Space Telescope Science Institute
Local Organizing Committee Member for The Metal-Poor Frontier: Understanding Low-Metallicity Stars and Galaxies Across Cosmic Time, 2026.

Johns Hopkins University
Physics and Astronomy Outreach Chair, 2023–2024.

Maria Mitchell Association
Observatory volunteer 2019–2025
Women of Science Symposium Advisory Committee Member, 2019–2022.

Mount Holyoke College
Teaching Assistant, 2018–2021.